

Convert Measurement Units

Lesson 4-6

Name: _____ Date: _____ Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.



units of length

1 foot = 12 inches, so the factor is 12

Conversion Factor

A ratio that helps you change one unit into another.

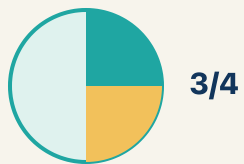


units of length

Inches, feet, pounds, ounces

Unit

A standard amount used to measure, like an inch or a liter.



$\frac{3}{4}$

2 feet = 24 inches

Equivalent

Having the same value, just in different units.



$3 : 2$

12 inches per 1 foot

Ratio

A way to compare two amounts.



units of length

*12 inches = 1 foot, 3 feet = 1 yard, 16 ounces = 1
pound*

Customary units

The U.S. way of measuring, like inches, feet, and pounds.



units of length

*1 meter = 100 centimeters, 1 kilogram = 1,000
grams*

Metric units

A way of measuring by 10s, like meters, liters, and grams.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can convert measurement units using ratios and conversion factors.

1 Notice

You're building a brand-new arcade cabinet!

2 Key idea

I can convert measurement units using ratios and conversion factors.

3 Words I'll use

Conversion Factor

Unit

Equivalent

Ratio

Customary units

Metric units



Think About It

- Why might the blueprint and the parts catalog use different units?
- How many inches are in 1 foot? How many feet in 1 yard?
- How do you decide whether to multiply or divide when converting?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: How many inches are in 7 feet?

- A. 84 inches
- B. 72 inches
- C. 70 inches
- D. 96 inches

Step 1

$$7 \times 12 = 84 \text{ inches.}$$



Answer: A. 84 inches

 We do — together

Solve this one with your class using the same steps.

Problem: How many feet are in 3 yards?

- A. 9 feet
- B. 6 feet
- C. 12 feet
- D. 36 feet

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: How many ounces are in 4 pounds?

- A. 64 ounces
- B. 48 ounces
- C. 32 ounces
- D. 16 ounces

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The cabinet blueprint uses feet but the parts catalog uses inches. Why do you have to convert, and which is the bigger unit, a foot or an inch?

Level 1 support

Start here: A foot is bigger than an inch, so I must convert to compare or combine the measurements.

Need a hint?

Sentence starters (optional)

I have to convert because the blueprint uses ____ but the catalog uses ____.

Tengo que convertir porque el plano usa ____ pero el catálogo usa ____.

One ____ is bigger than one ____.

Un/Una ____ es más grande que un/una ____.

Word bank: unit conversion factor customary units equivalent

Listen for: Students explain a foot is the larger customary unit (12 inches) and that mixed units must be converted to the same unit before working with them.

Level 2

**Predict: when you change feet into inches, will the number get bigger or smaller?
Explain using the size of each unit.**

- The number gets ____ because ____.
- Switching to a smaller unit means ____.

EXPLORE

How did you use a conversion factor like '1 ft = 12 in' to fill in the ratio table, and how did you decide whether to multiply or divide?

Level 1 support

Start here: I multiply when going to a smaller unit and divide when going to a bigger unit.



Need a hint?

Sentence starters (optional)

To convert ____ **to** ____, **I** ____ **by** ____.

Para convertir ____ *a* ____, ____ *por* ____.

____ **equals** ____ **because the conversion factor is** ____.

____ *es igual a* ____ *porque el factor de conversión es* ____.

Word bank: (conversion factor) (unit) (equivalent) (ratio) (customary units)

Listen for: Students use the conversion factor as a ratio, multiply going to smaller units ($5 \text{ ft} \times 12 = 60 \text{ in}$) and divide going to larger units ($48 \text{ in} / 12 = 4 \text{ ft}$).

Level 2

Feet-to-inches uses multiply but inches-to-feet uses divide, even though both use '12.'

Explain why the same conversion factor leads to opposite operations.

- I multiply when ____ but divide when ____.
- The same factor works both ways because ____.

CONNECT

A doorway is 80 inches tall and a new cabinet is listed as 7 feet. Your friend says it won't fit because 7 feet is 82 inches. Will it fit, and where is the error?

Level 1 support

Start here: 7 feet is 84 inches, which is taller than the 80-inch doorway, so it will not fit.

 **Need a hint?**

Sentence starters (optional)

7 feet = 7 x ____ = ____ inches.

7 pies = 7 x ____ = ____ pulgadas.

Since ____ inches is ____ than 80 inches, the cabinet ____ fit.

Como ____ pulgadas es ____ que 80 pulgadas, el gabinete ____ caber.

Word bank: conversion factor unit equivalent customary units

Listen for: Students convert 7 ft to 84 in (not 82), compare $84 > 80$, and conclude the cabinet will not fit; they catch the friend's arithmetic error.

Level 2

How many inches would you need to trim off the cabinet to make it fit, and is that even possible for a real cabinet? Justify your reasoning.

- You would need to remove ____ inches because ____.
- That is or is not realistic because ____.

Write About the Math

The Writing Revolution

I can explain my conversion using the words conversion factor, unit, equivalent, and ratio.

1. Kernel Sentence subject + verb

Model: Unit is a standard amount used to measure, like an inch or a liter.

Unidad es una cantidad estándar para medir, como una pulgada o un litro.

Write a kernel sentence about unit. Use a subject and a verb.

Escribe una oración base sobre unidad. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Unit matters in math

Unidad importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Unit matters in math because ____.

Unidad importa en matemáticas porque ____.

but
pero

Unit matters in math, but ____.

Unidad importa en matemáticas, pero ____.

so
entonces

Unit matters in math, so ____.

Unidad importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about unit.
Di un hecho verdadero sobre unit.

Unit ____.

Question *Pregunta*

Ask a question about unit.
Haz una pregunta sobre unit.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about unit.
Muestra entusiasmo sobre unit.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with unit.
Dile a un compañero qué hacer con unit.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I have to convert because the blueprint uses ____ but the catalog uses ____.

Tengo que convertir porque el plano usa ____ pero el catálogo usa ____.

One ____ is bigger than one ____.

Un/Una ____ es más grande que un/una ____.

To convert ____ to ____, I ____ by ____.

Para convertir ____ a ____, ____ por ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. A recipe needs 3 quarts of water. How many cups is that? (1 qt = 4 cups)

- A. 12
- B. 7
- C. 8
- D. 16

Show your work:

2. How many pounds are in 64 ounces? (1 lb = 16 oz)

- A. 4
- B. 8
- C. 48
- D. 1,024

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A recipe says you need 5 feet of ribbon, but the store only sells ribbon in inches. Another recipe says you need 2 yards of fabric, but your ruler measures in feet. Convert both measurements and explain the steps you used. Which conversion factor did you use each time?

Sentence starter: 5 feet = ____ inches because ____ . 2 yards = ____ feet because ____ . The conversion factors I used were ____ .

Show your work:

Reflect — Exit Ticket

Which is the correct conversion of 5 yards to inches?

- A. 180 inches
- B. 60 inches
- C. 15 inches
- D. 540 inches

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 9 feet — $1 \text{ yard} = 3 \text{ feet}$. $3 \times 3 = 9 \text{ feet}$.
2. **You Try (notes):** A. 64 ounces — $1 \text{ pound} = 16 \text{ ounces}$. $4 \times 16 = 64 \text{ ounces}$.
3. **Try It 1:** A. $12 - 3 \times 4 = 12 \text{ cups}$.
4. **Try It 2:** A. $4 - 64 \div 16 = 4 \text{ pounds}$.
5. **Exit Ticket:** A. 180 inches — $5 \text{ yards} \times 3 = 15 \text{ feet}$. $15 \text{ feet} \times 12 = 180 \text{ inches}$. Or: $5 \times 36 = 180 \text{ inches}$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Unit is a standard amount used to measure, like an inch or a liter.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).